

Task 2: Deep Learning — Handwritten Digit Recognition

Objective

The objective of this project is to build and evaluate a deep learning model capable of recognizing handwritten digits (0–9) using the MNIST dataset.

Tools & Technologies

Python 3.x, TensorFlow/Keras, NumPy, Matplotlib, Google Colab

Dataset

Training images: 60,000

Testing images: 10,000

Image size: 28 × 28 pixels

Data Preprocessing

1. Loaded dataset using TensorFlow
2. Visualized sample digit images
3. Normalized pixel values from 0–255 to 0–1

Model Architecture

Flatten Layer

Dense Hidden Layer (128 neurons, ReLU)

Output Layer (10 neurons, Softmax)

Total Parameters: 101,770

Model Training

Optimizer: Adam

Loss Function: Sparse Categorical Crossentropy

Metric: Accuracy

Epochs: 5

Training Accuracy: 90.93%

Test Accuracy: 91.94%

Prediction Example

Actual Label: 7

Predicted Label: 7

Conclusion

The deep learning model successfully classified handwritten digits using the MNIST dataset with an accuracy of 91.94%.